

Ashutosh Madhu

Software Professional | UX Designer & Backend Engineer
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SUMMARY

Mathematics–Computer Science graduate from UC San Diego with hands-on experience in software engineering, backend development, API design, software testing, debugging, and data processing. Proficient in Python, Java, C++, and SQL, with academic experience in data structures, algorithms, databases, and software development. Built a Python/FastAPI environmental data API involving data retrieval, processing, normalization, and modular backend design. Experienced in investigating software issues, developing technical solutions, and collaborating in technology-focused projects. Interested in building scalable software and learning new technologies to solve complex engineering problems.

EDUCATION

University of California, San Diego (UC San Diego)
B.S. Mathematics–Computer Science | Expected Summer 2026

PROJECTS

Project-Oxy — Environmental Data API | Feb 2026 - August 2026 | San Diego
Python | FastAPI | Backend Development | API Design | Data Processing

- Built a Python/FastAPI backend that retrieves and processes country-level environmental and air-quality data.
- Designed API endpoints for country lookup, air-quality data, pollution scoring, oxygen scoring, and eco-balance estimates.
- Organized backend functionality into modular components for data retrieval, country data, and scoring logic.
- Applied software engineering concepts including API design, data normalization, debugging, and structured backend development.
- Investigated application behavior and refined backend logic while developing and testing API functionality.
- Translated environmental data requirements into programmatic logic and usable API endpoints.

PROFESSIONAL EXPERIENCE

Software Tester — TestIO / EPAM | Jan 2025 - December 2025 | San Diego

- Tested web and mobile applications by investigating functional defects, unexpected behavior, and usability issues.
- Reproduced software issues and documented clear steps, expected results, and actual results to support defect resolution.

- Analyzed application behavior from a user's perspective and identified inconsistencies requiring further investigation.
- Developed practical experience with software quality, debugging, defect analysis, and testing throughout the software development process.

CompTechS Intern | July 2024 - December 2024 | San Diego

- Diagnosed hardware and software problems on laptops and desktop computers and identified appropriate solutions.
- Replaced hardware components and reinstalled required software to restore systems to working condition.
- Investigated performance and configuration issues before devices were distributed to students.
- Developed systematic troubleshooting and problem-solving skills while working with varied technical issues.

LEADERSHIP & COMMUNITY

Kolthuruthi Government School | July 2020 | Kannur, Kerala, India

- Helped lead a technology initiative to raise funds for laptops for a rural school.
- Supported the implementation of technology resources designed to improve students' access to computing and educational software.
- Installed free educational software and helped prepare computers for student use.
- Contributed to a community-focused technology project requiring coordination, problem solving, and hands-on technical implementation.

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL

Frameworks & Tools: FastAPI, Git, GitHub, Microsoft Excel

Software Engineering: Data Structures, Algorithms, Object-Oriented Programming, API Development, Backend Development, Software Testing, Debugging, Data Processing, Data Normalization, Database Fundamentals

ADDITIONAL

Robotics Programming | Debate | ABRSM Piano/Music Theory | Martial Arts | Team Athletics